



Analysis of Developers Survey

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OUTLINE



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EXECUTIVE SUMMARY



- The vast majority of developers were male accounting for 92.5%.
- The top language used by developers was HTML/CSS.
- The top databased utilized by developers was MySQL.
- The most widely used web frame was jQuery.
- Docker is the most used platform.
- The main age was between 20 and 40.
- The majority of respondents were holders of bachelor's degrees.
- USA was a key country for the most developers.

INTRODUCTION



- **Overview of the Report:** This report presents the results of a comprehensive survey conducted among programmers and developers worldwide, primarily focusing on their experiences, preferences, and the technologies they use.
- **Target Audience:** The report is tailored for stakeholders in the tech industry, including software engineers, data scientists, IT managers, and decision-makers looking to understand current trends in technology and developer preferences.

INTRODUCTION



- **What You'll Learn:** By going through this report, readers will gain insights into the most popular programming languages, the average salaries associated with different technologies, and emerging trends in developer tools and technologies.
- The analysis aims to help readers make informed decisions regarding tech stacks, career paths, and strategies for talent acquisition and retention.

METHODOLOGY



- Jupyter notebook was used as platform for data analysis.
- Google Looker Studio was used to create dashboards.
- Python was used for data analysis and visualizations.
- Median was used because the data is right-skewed.
 - Correlation factors among all variables were calculated.
- Data wrangling techniques were applied to find missing values and to clean data from outliers.
- The data was acquired from a survey from StackOverflow Developers.

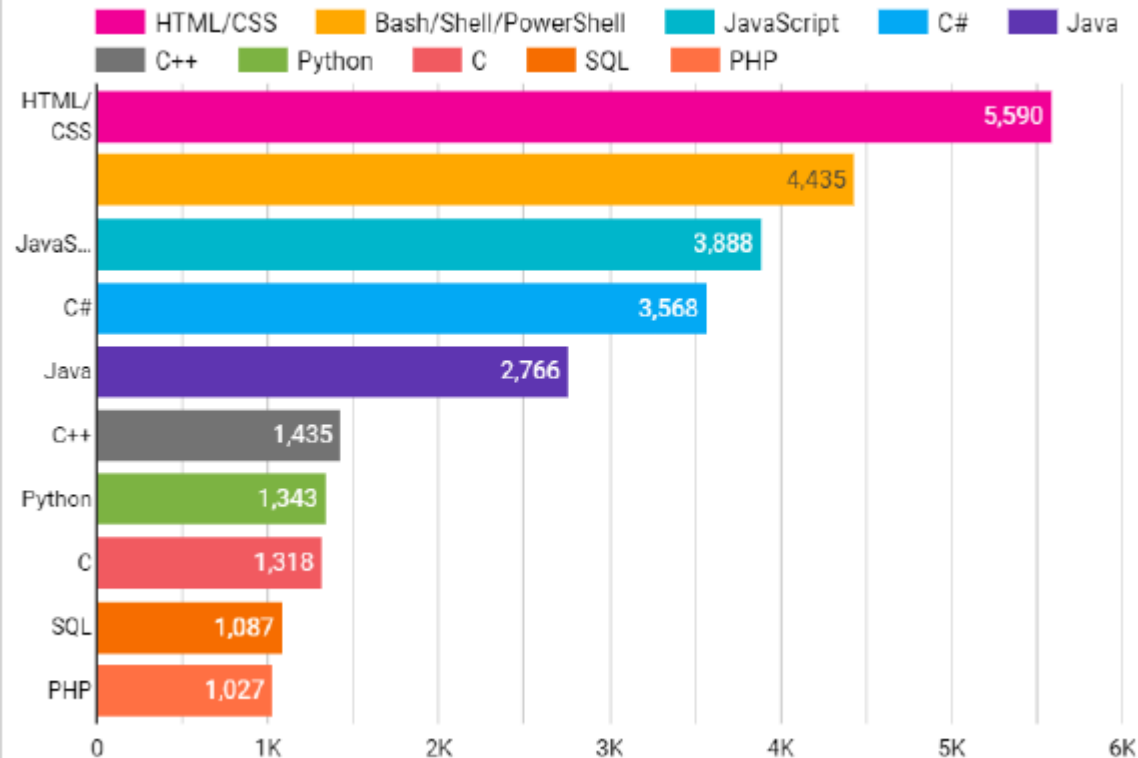
RESULTS



PROGRAMMING LANGUAGE TRENDS

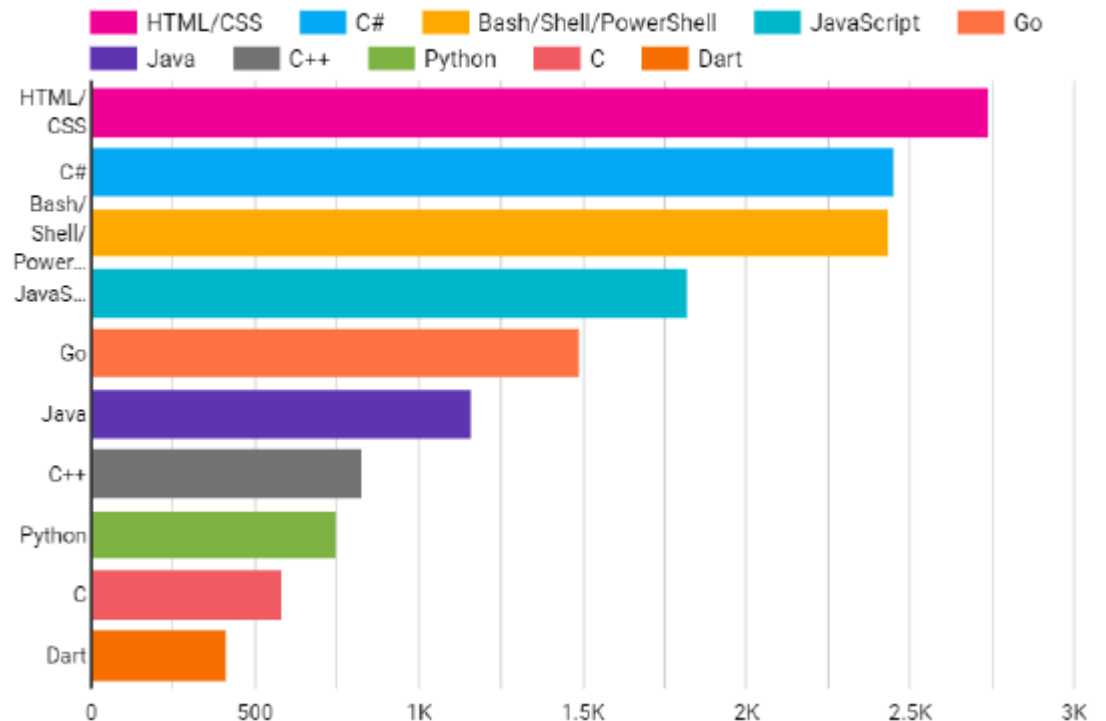
Current Year

Top 10 Languages Worked With



Next Year

Top 10 Desired Language for Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

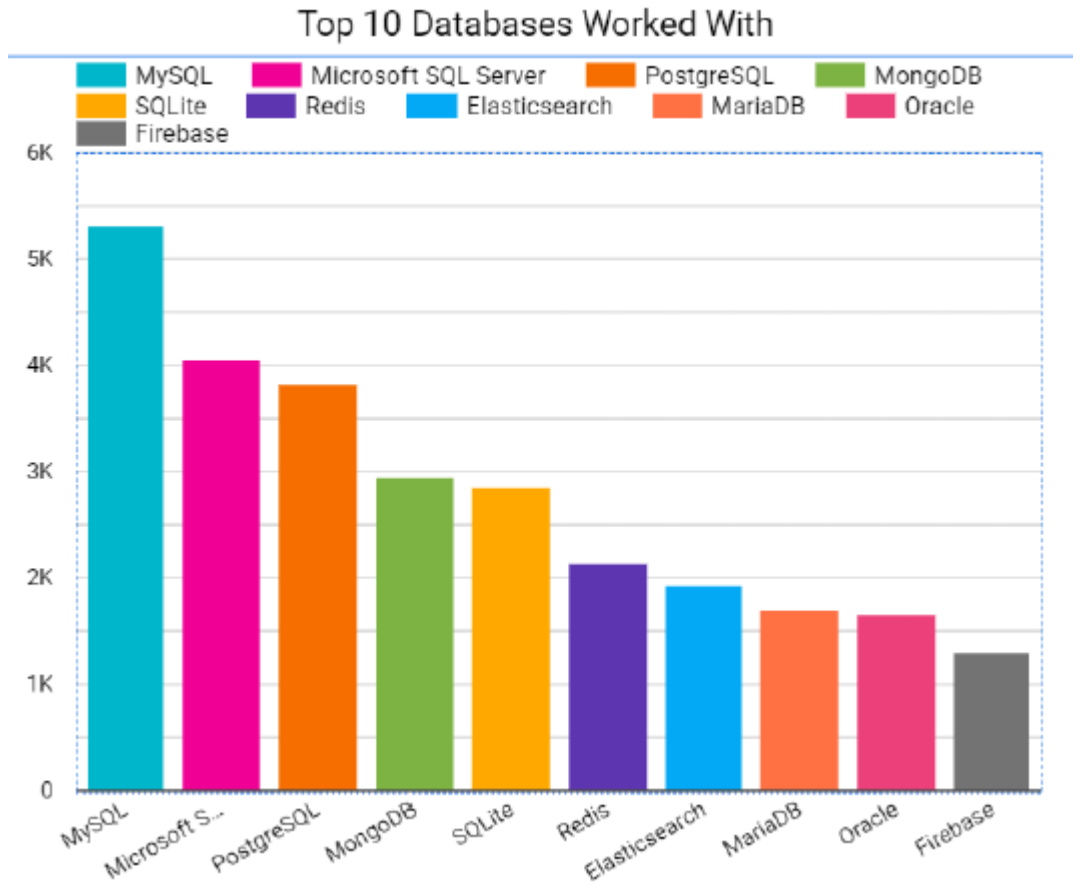
- HTML/CSS was found to be the top used and desired language for developers.
- Bash/Shell/PowerShell ranked the second most widely used language and the third desired language.
- PHP was the least utilized while Dart was the least desired in the ranking.

Implications

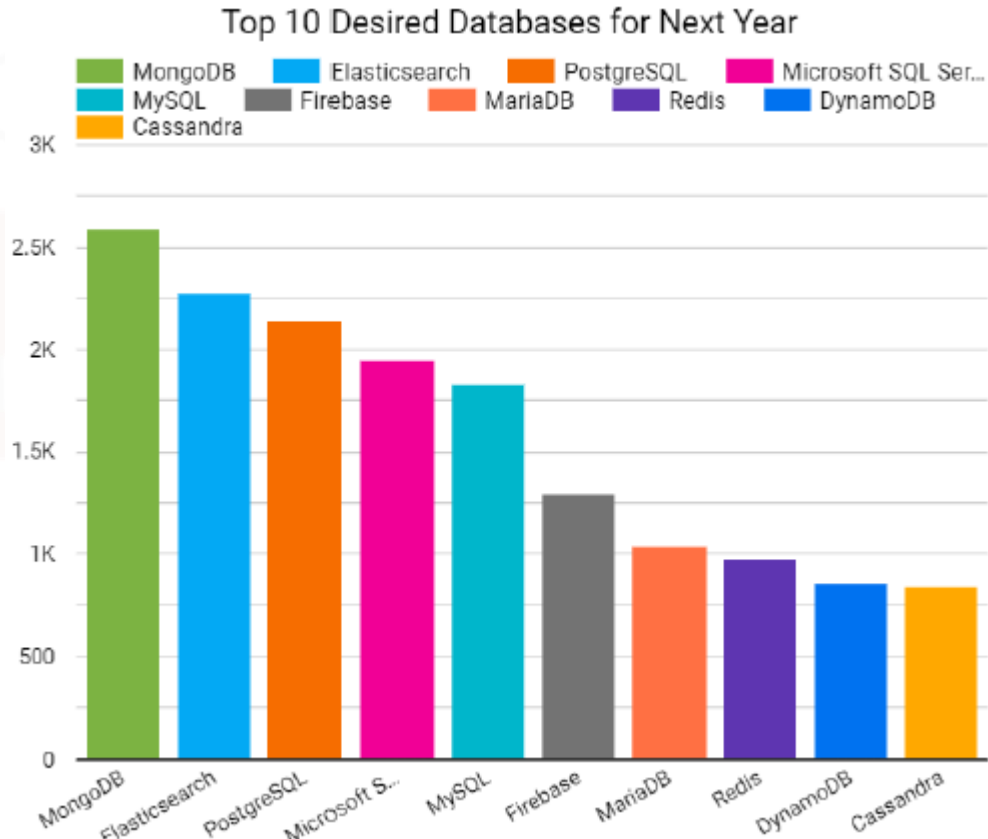
- Scripting languages such as Bash/Shell/PowerShell are viral and have a huge supportive community and a huge learning community.
- Web application depending on HTML/CSS are the most popular.
- C# from Microsoft is wanted among developers and has a huge supportive community.

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS

- **MongoDB's Rise:** MongoDB has experienced a significant increase in both usage and desire, suggesting a growing preference for NoSQL databases for modern applications.
- **MySQL's Persistence:** Despite the rise of NoSQL, MySQL remains the most widely used database, indicating its continued reliability and value in established environments.
- **Shift Towards Cloud-Based Solutions:** Cloud-based databases like Firebase and Microsoft SQL Server have gained popularity, reflecting a trend towards cloud adoption in database management.

DATABASE TRENDS - IMPLICATIONS

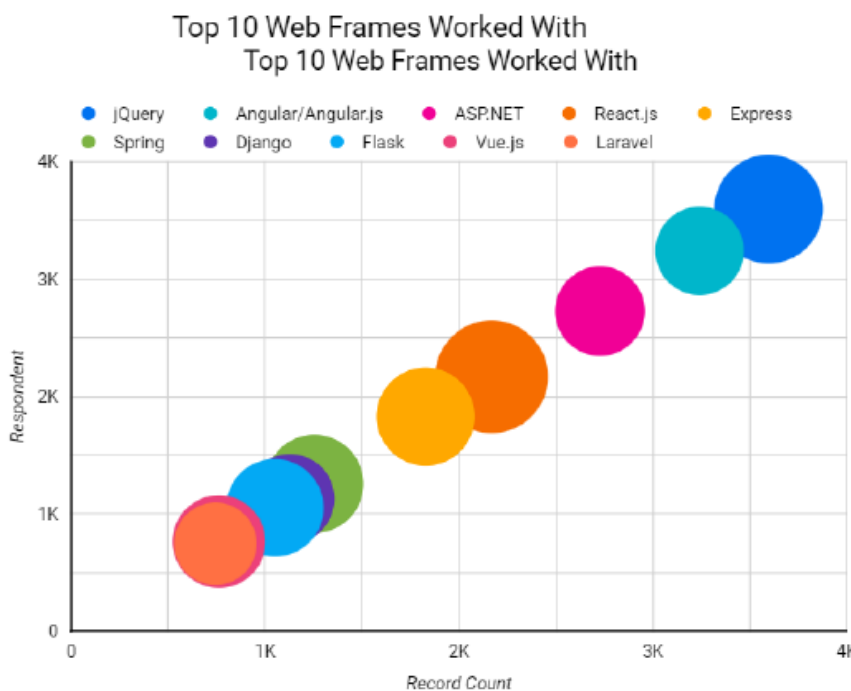
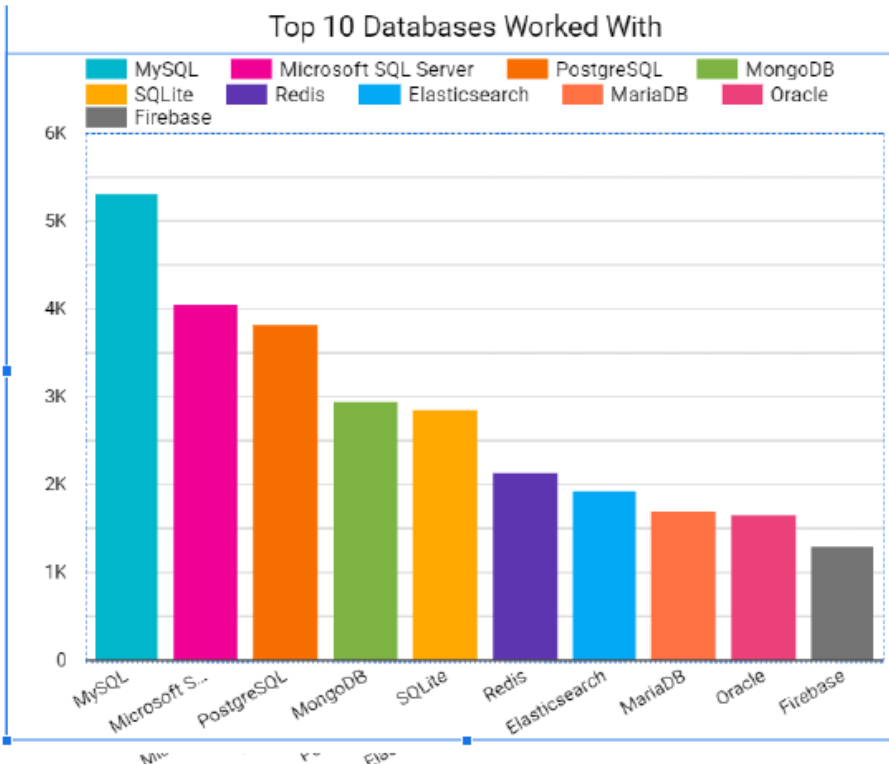
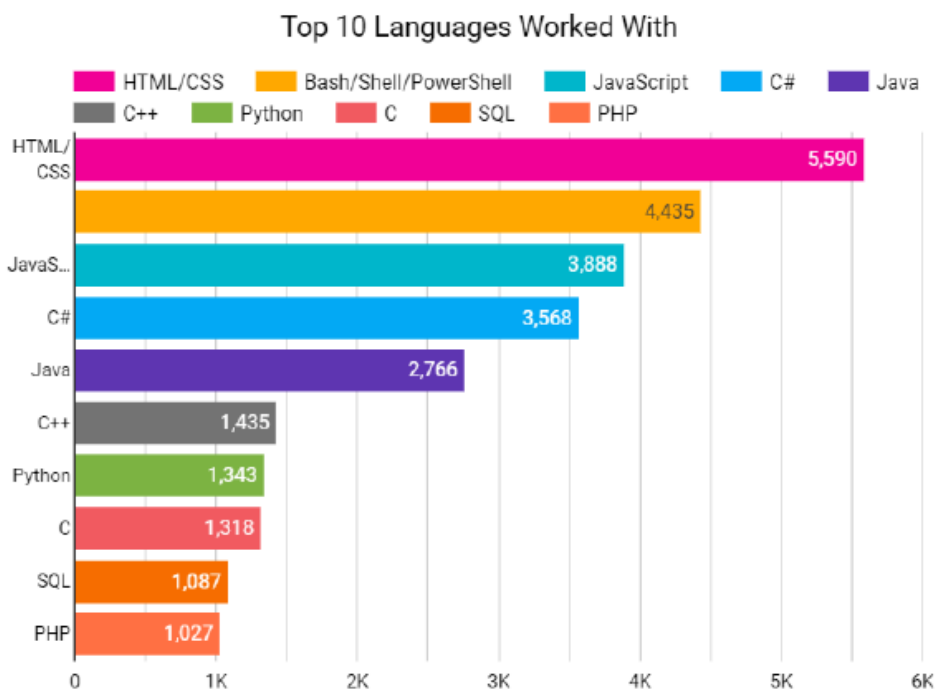
- **Technology Adoption:** Organizations should consider adopting MongoDB and other NoSQL databases to meet the evolving needs of modern applications, particularly those requiring flexibility and scalability.
- **Skill Development:** Developers should focus on acquiring skills in MongoDB, NoSQL databases, and cloud-based database technologies to stay competitive in the job market.
- **Database Selection:** The choice of database should be carefully evaluated based on specific project requirements, considering factors like data volume, scalability needs, performance, and integration with other tools.

DASHBOARD



<https://github.com/a7madv4d2/Dashboard>

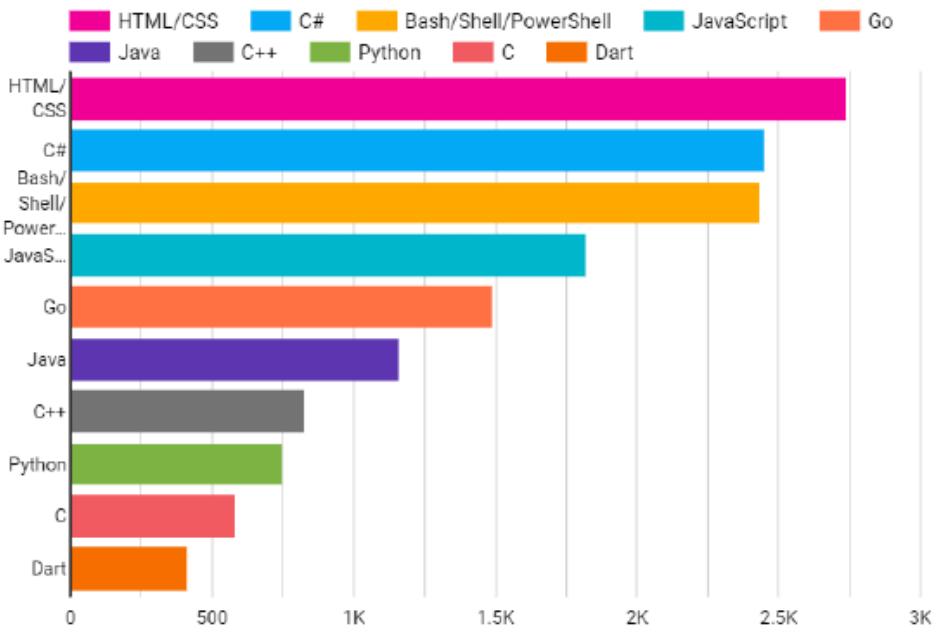
DASHBOARD TAB 1



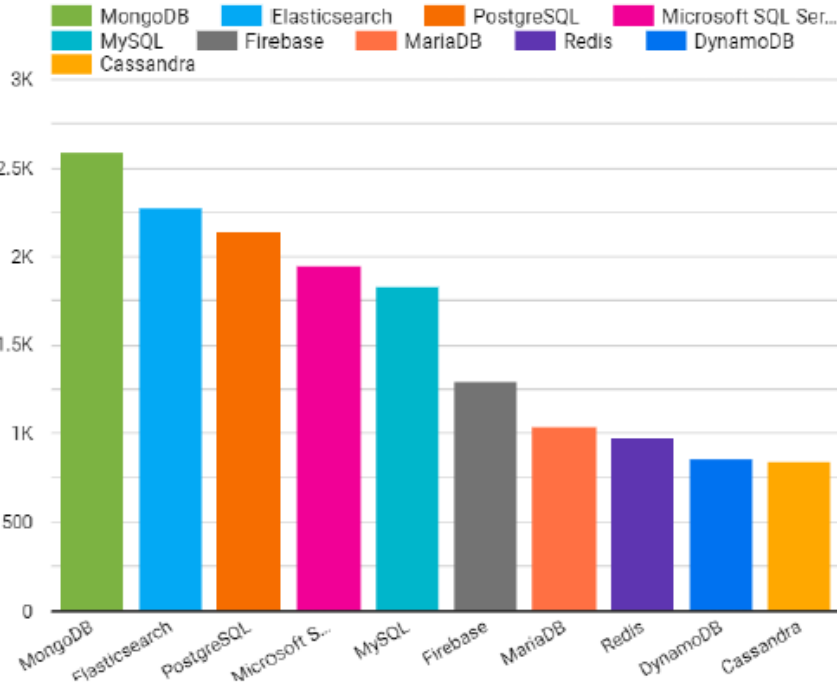
DASHBOARD TAB 2

IBM Developer

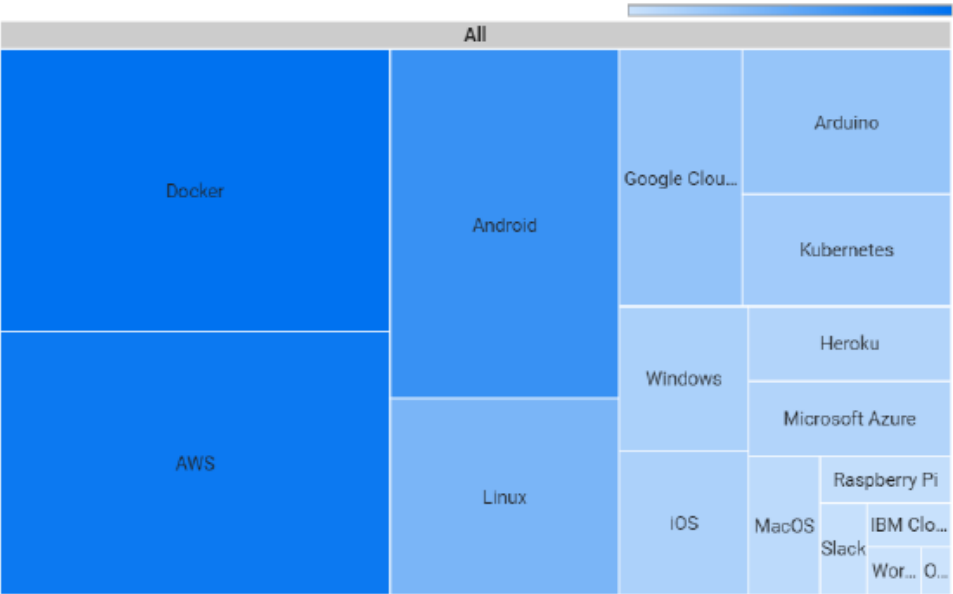
Top 10 Desired Language for Next Year



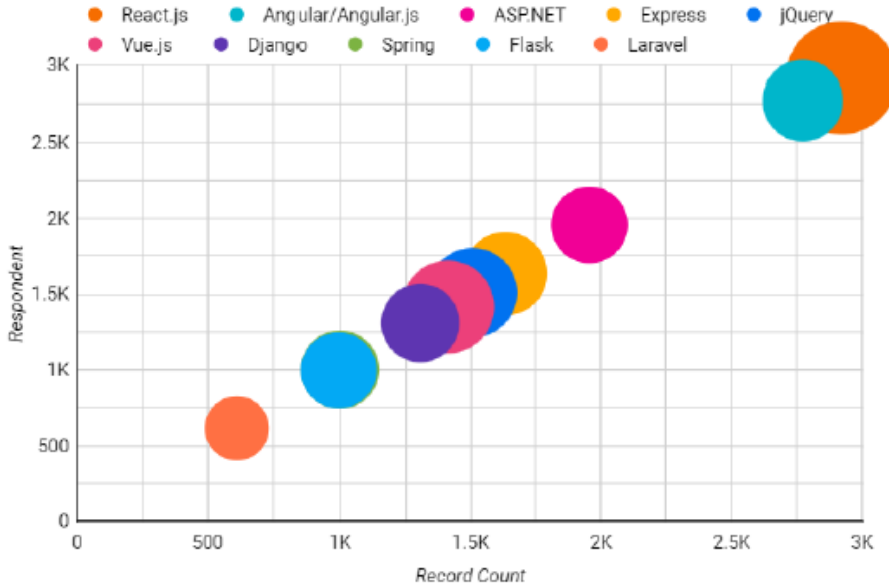
Top 10 Desired Databases for Next Year



Top 10 Desired Platforms for Next Year



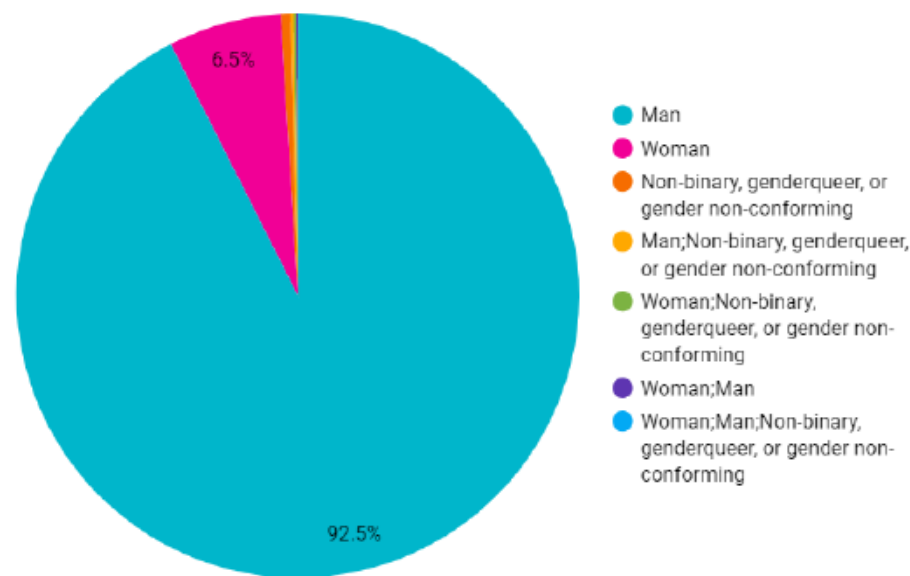
Top 10 Desired Web Frameworks for Next Year



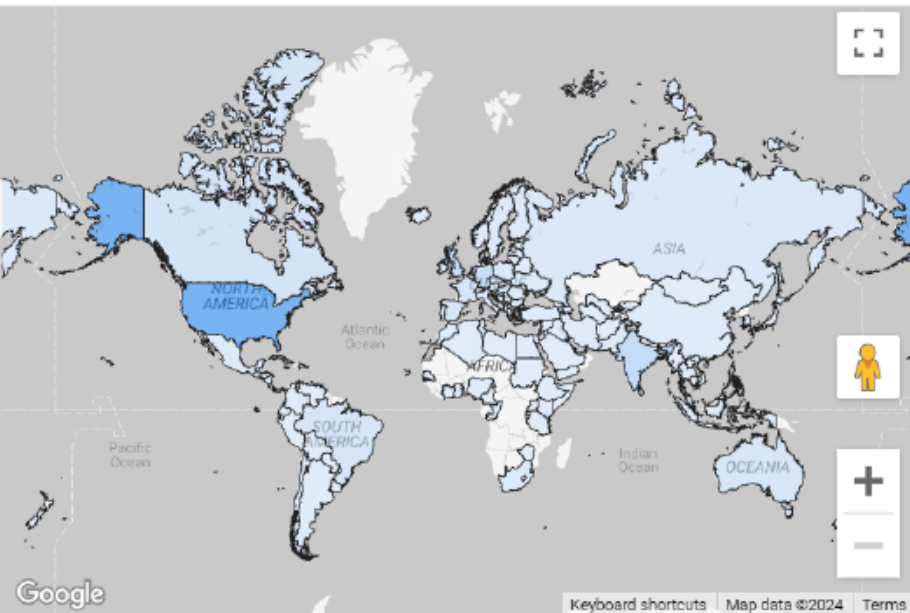
DASHBOARD TAB 3

IBM Develope

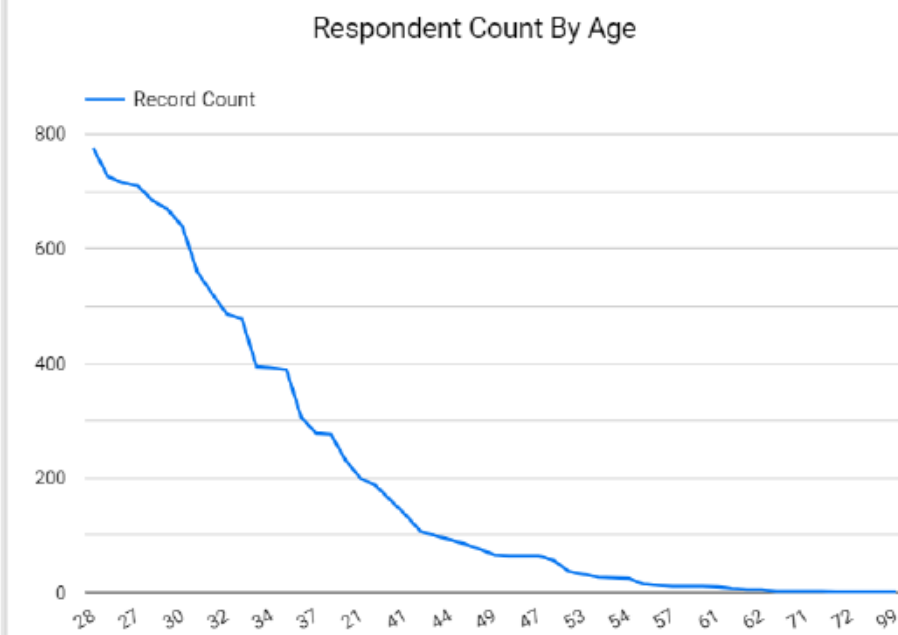
Respondent classified by Gender



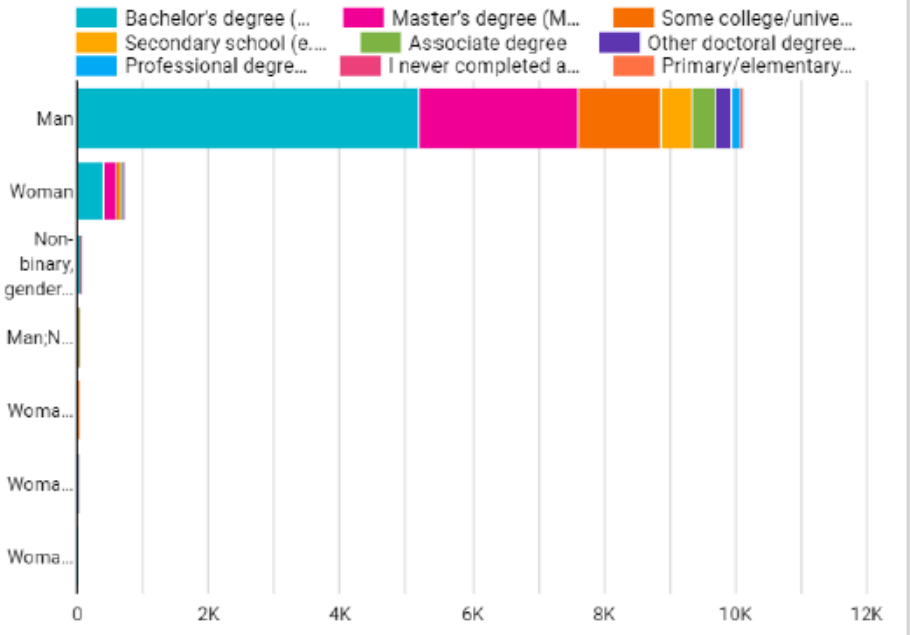
Respondent Count for Countries



Respondent Count By Age



Record Count 1 3,019



DISCUSSION



OVERALL FINDINGS

Shift Towards Modern Technologies: The increasing popularity of JavaScript, Docker, and MongoDB suggests a shift towards modern programming languages and databases that are well-suited for building scalable and dynamic applications.

Diverse Technology Stack: Developers are using a diverse range of programming languages and databases, reflecting the complexity and variety of modern software development projects.

OVERALL FINDINGS

Consistent Language Preferences: The top desired languages for the next year remain similar to the current year, with JavaScript, and C# leading the list.

Database Diversity: While MySQL and PostgreSQL remain popular, there is a growing interest in NoSQL databases like MongoDB and Elasticsearch, as well as cloud-based solutions like Firebase.

Cloud Platform Popularity: Docker, Azure, and AWS continue to be the most desired cloud platforms, reflecting the growing adoption of cloud computing.

Web Framework Variety: A wide range of web frameworks are desired, with React, Angular, and Express leading the list, indicating the diversity of web development approaches.

OVERALL FINDINGS

Gender Diversity: The respondent pool is predominantly **male**, with a smaller percentage of women and non-binary individuals.

Age Distribution: The majority of respondents are **between 25 and 44** years old, indicating a focus on mid-career professionals.

Educational Background: Most respondents have **at least a bachelor's degree**, highlighting the importance of education in the tech industry.

Geographic Distribution: The map likely shows the distribution of respondents across different countries, providing insights into regional trends and variations. The major respondents were from the U.S.

OVERALL IMPLICATIONS

- **Skill Development:** Developers should focus on acquiring skills in JavaScript, Python, and MongoDB to stay competitive in the job market.
- **Technology Adoption:** Organizations should consider adopting modern technologies like JavaScript, Python, and MongoDB to build innovative and scalable applications.
- **Database Selection:** The choice of database should be carefully evaluated based on specific project requirements, considering factors like data volume, scalability needs, and performance.

OVERALL IMPLICATIONS

- **Diversity and Inclusion:** Organizations should prioritize initiatives to promote gender diversity and inclusion in the tech industry.
- **Talent Acquisition:** Companies should focus on attracting and retaining experienced professionals with strong educational backgrounds.
- **Global Reach:** The geographical distribution of respondents can inform strategies for expanding business operations and targeting specific markets.

CONCLUSION



- **Shift Towards Modern Technologies:** The increasing popularity of JavaScript, Python, NoSQL databases, and cloud-based solutions reflects a shift towards modern development practices.
- **Full-Stack Development:** The combination of programming languages, databases, frameworks, and platforms indicates a growing demand for full-stack developers.
- **Cloud-Native Adoption:** Cloud-native technologies and architectures are becoming more prevalent, driven by the benefits of scalability, flexibility, and efficiency.
- **Diversity and Inclusion:** Addressing the gender gap and promoting diversity and inclusion in the tech industry is crucial for fostering innovation and growth.

CONCLUSION



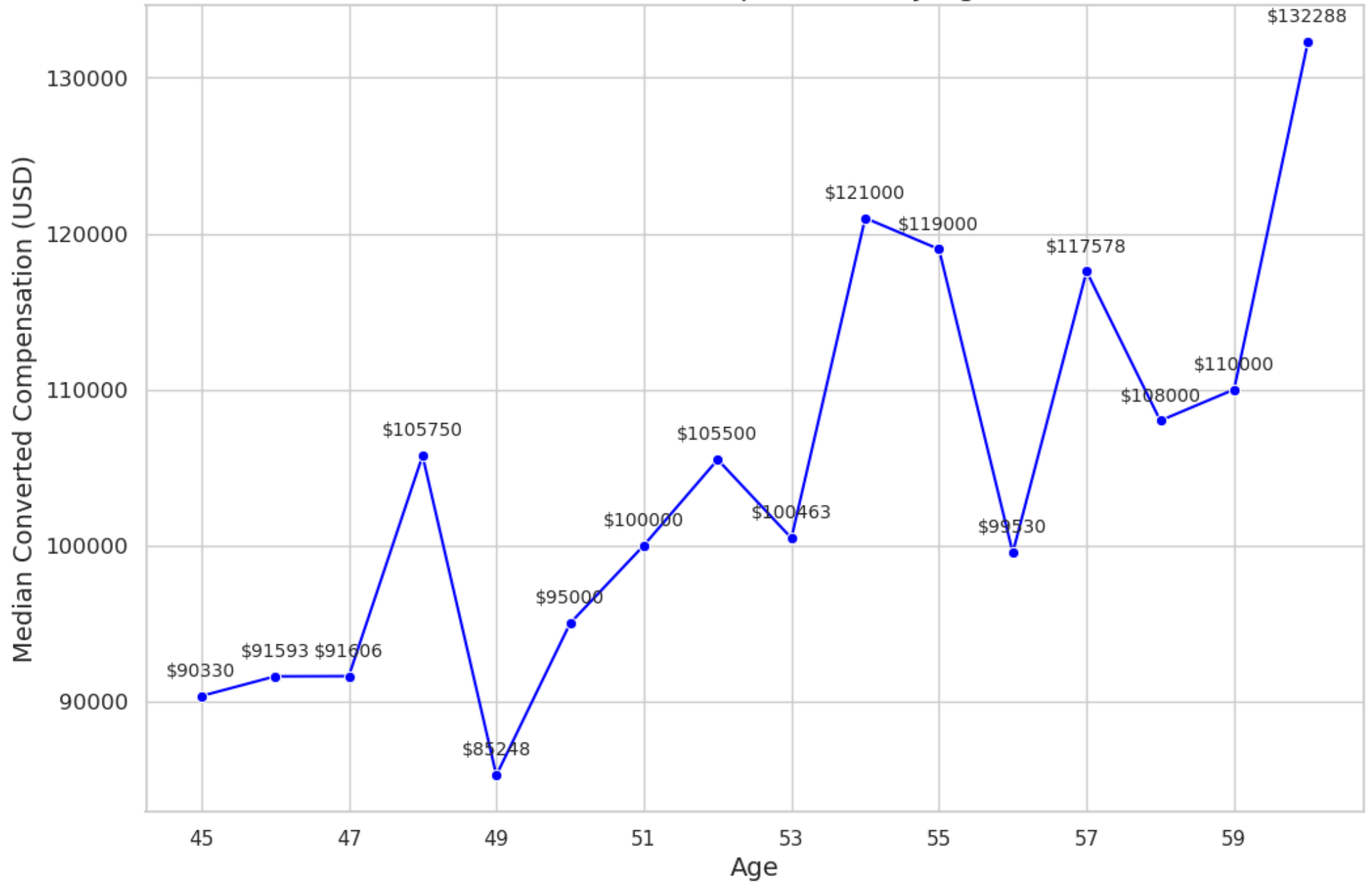
Demographic Trends

- **Gender Gap:** The data suggests a gender gap in the tech industry, with a higher proportion of male respondents.
- **Experienced Workforce:** The majority of respondents are mid-career professionals, indicating a mature and experienced workforce.
- **Educational Background:** Education plays a significant role, with most respondents holding at least a bachelor's degree.

APPENDIX

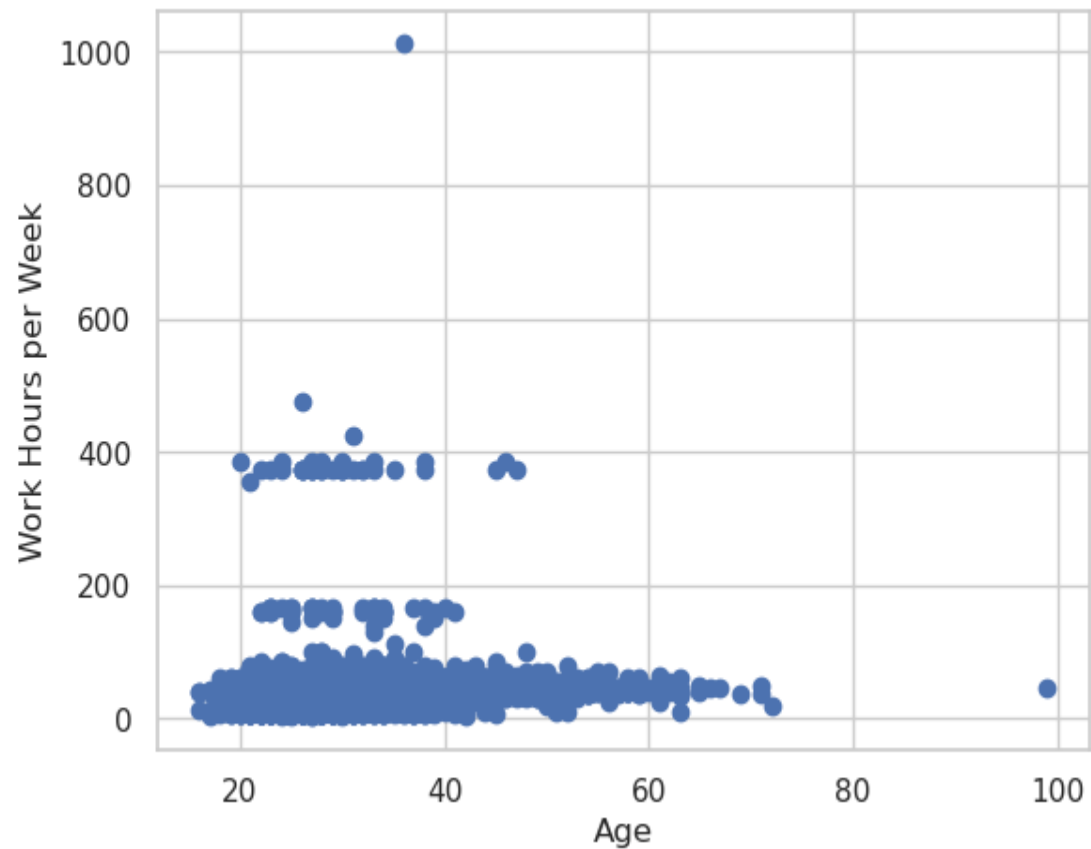


Median Converted Compensation by Age (45-60)

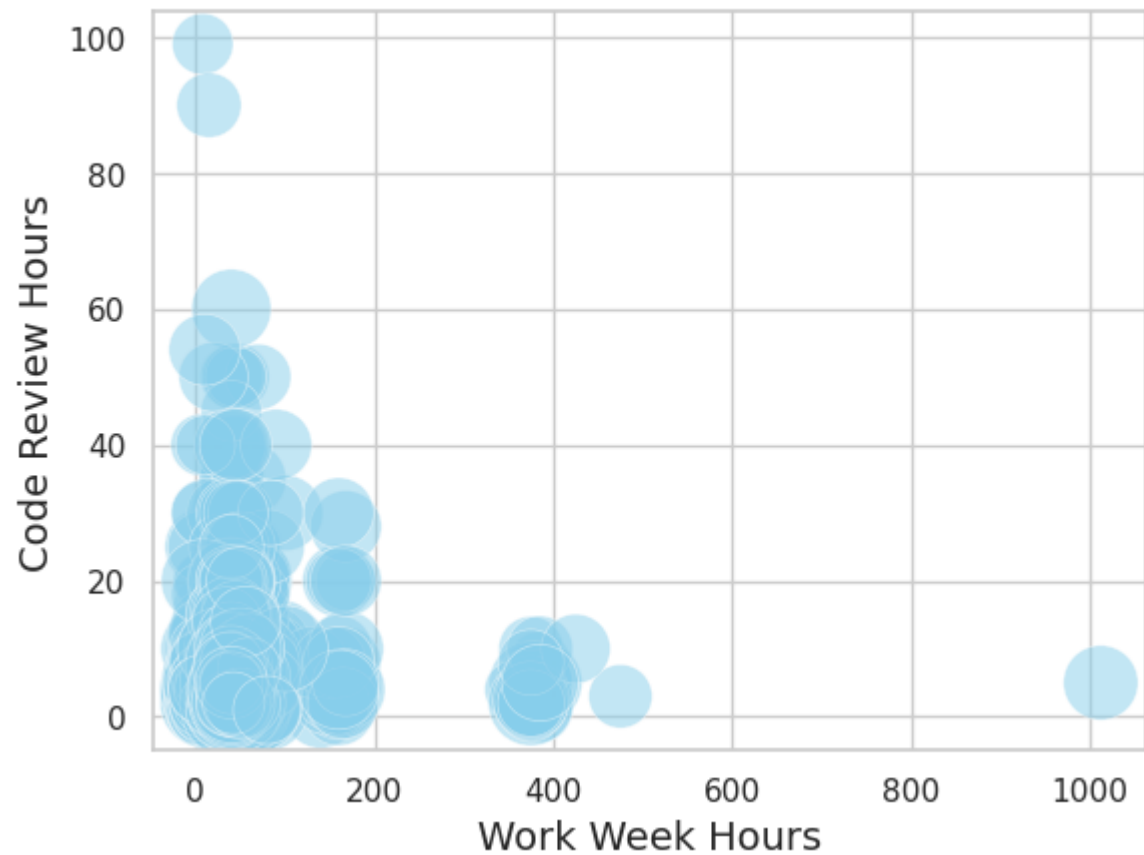


Median Work Week Hours and Code Review Hours for Age Group 30-35

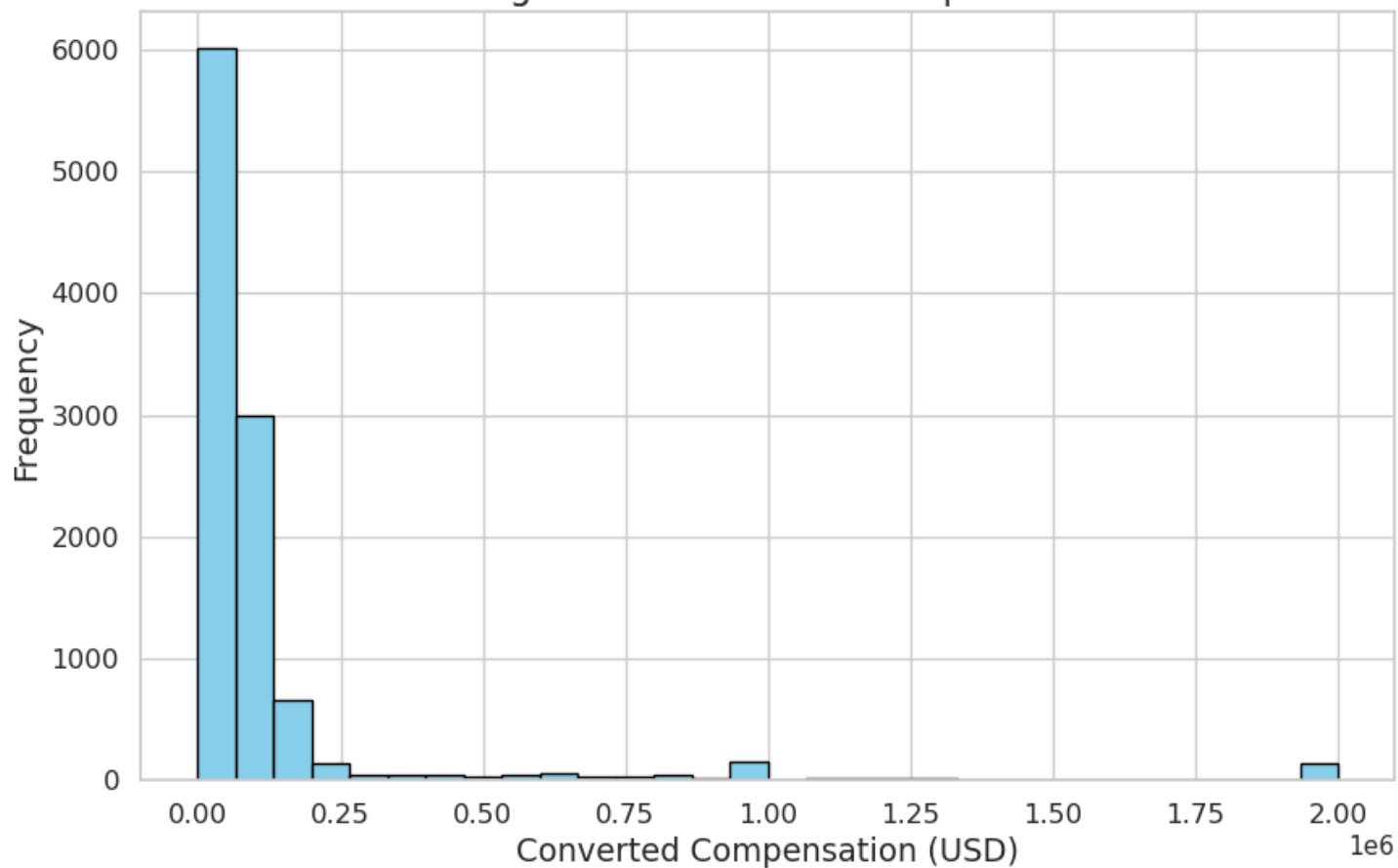




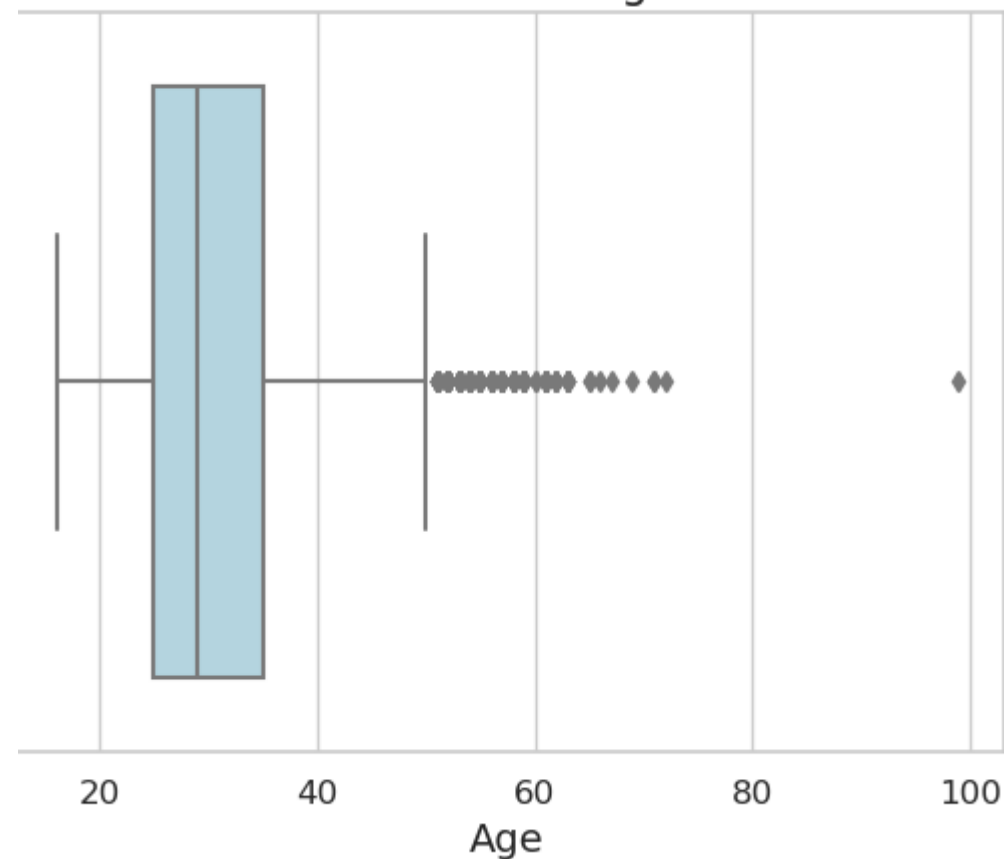
Bubble Plot of Work Week Hours vs. Code Review Hours



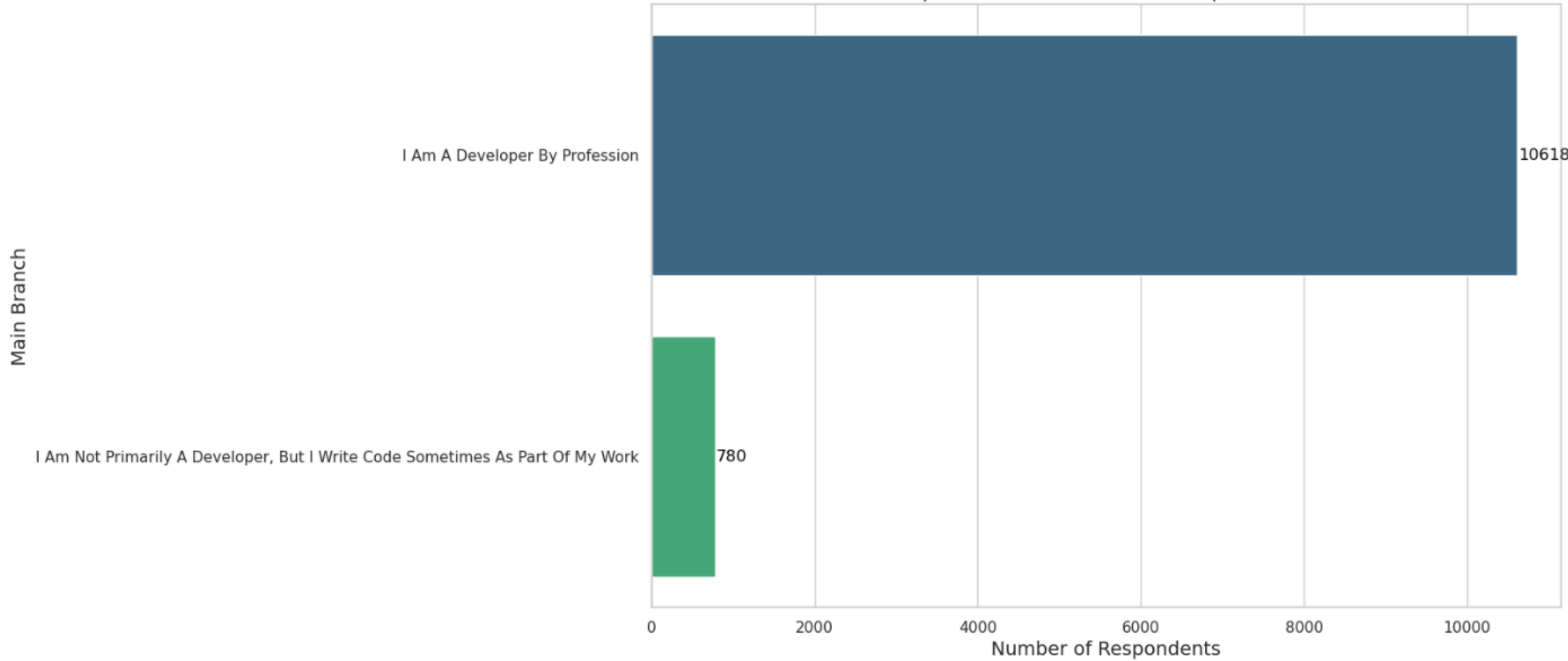
Histogram of Converted Compensation



Box Plot of Age

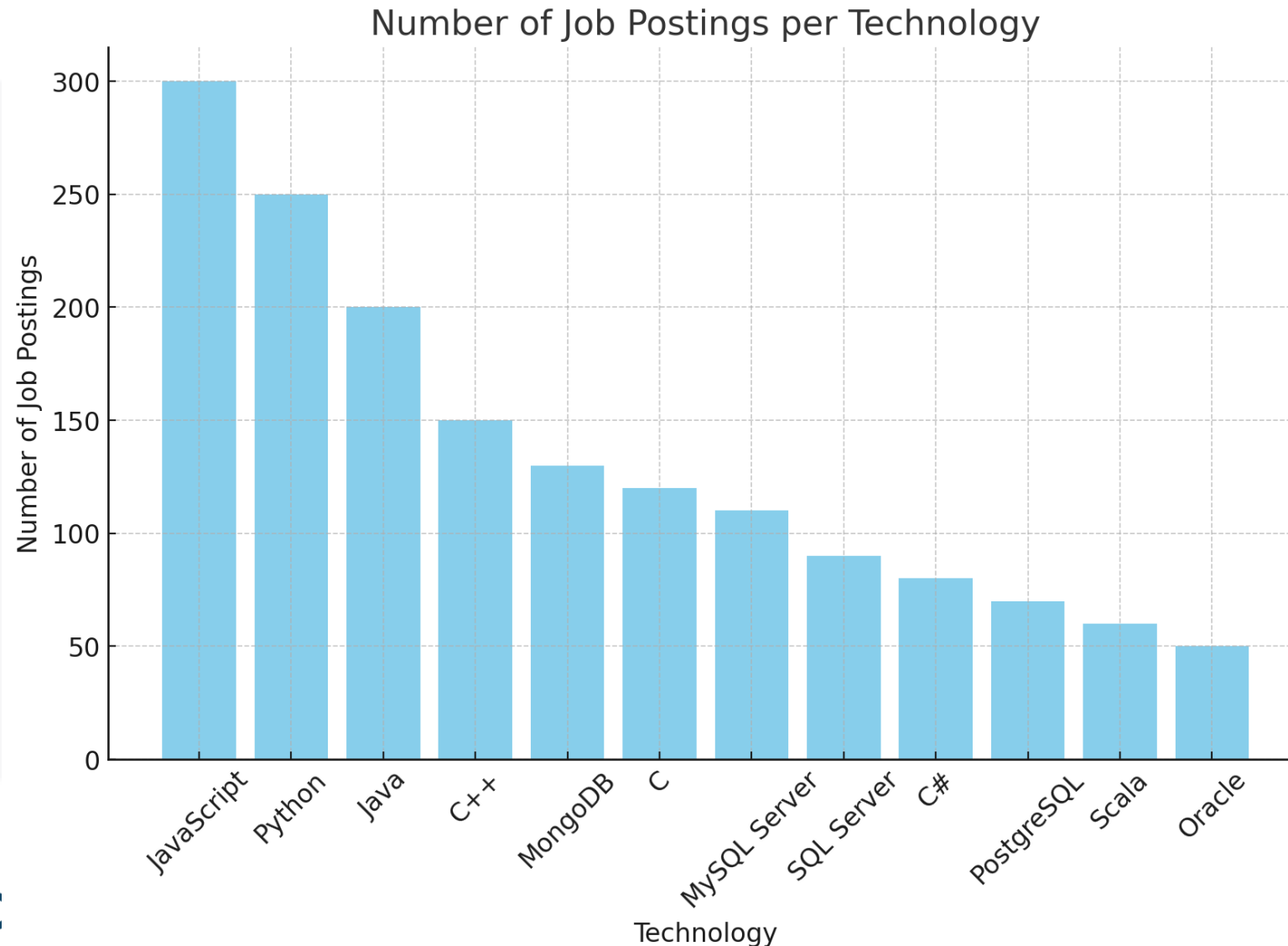


Top 10 Main Branches of Respondents



	Respondent	CompTotal	ConvertedComp	WorkWeekHrs	CodeRevHrs	Age	LogConvertedComp
Respondent	1.000000	-0.013482	0.002181	-0.016178	0.007031	0.002394	0.008563
CompTotal	-0.013482	1.000000	0.001037	0.003467	0.006698	0.006949	-0.020942
ConvertedComp	0.002181	0.001037	1.000000	0.021143	-0.033865	0.105386	0.633313
WorkWeekHrs	-0.016178	0.003467	0.021143	1.000000	0.032463	0.031592	0.044200
CodeRevHrs	0.007031	0.006698	-0.033865	0.032463	1.000000	-0.015742	-0.124717
Age	0.002394	0.006949	0.105386	0.031592	-0.015742	1.000000	0.292882
LogConvertedComp	0.008563	-0.020942	0.633313	0.044200	-0.124717	0.292882	1.000000

JOB POSTINGS



POPULAR LANGUAGES

